

**Ans.** i) Total number of trials = 1000.

The frequency of a tyre required to be replaced before covering 4000 km  
= 20

$$\text{So, } P(E_1) = 20/1000 = 0.02$$

ii) The frequency that tyre will last more than 9000 km = 325 + 445 = 770

$$\text{So, } P(E_2) = 770/1000 = 0.77$$

iii) The frequency that tyre requires replacement between 4000 km and 14000 km = 210 + 325 = 535.

$$\text{So, } P(E_3) = 535/1000 = 0.535$$

**Q4.** A coin is tossed three times, consider the following events.

P: 'No head appears',

Q: 'Exactly one head appears' and

R: 'At Least two heads appear'.

Check whether they form a set of mutually exclusive and exhaustive events.

**Ans:** The sample space of the experiment is:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\} \text{ and}$$

$$P = \{TTT\},$$

$$Q = \{HTT, THT, TTH\},$$

$$R = \{HHT, HTH, THH, HHH\}$$

$$P \cup Q \cup R = \{TTT, HTT, THT, TTH, HHT, HTH, THH, HHH\} = S$$

Therefore, P, Q and R are exhaustive events.

and

$$P \cap Q = \varnothing, P \cap R = \varnothing \text{ and } Q \cap R = \varnothing$$

Therefore, the events are mutually exclusive.

Hence, P, Q and R form a set of mutually exclusive and exhaustive events.

**Q5.** If  $P(A) = 7/13$ ,  $P(B) = 9/13$  and  $P(A \cap B) = 4/13$ , evaluate  $P(A|B)$ .

$$\text{Ans: } P(A|B) = P(A \cap B)/P(B) = (4/13)/(9/13) = 4/9$$

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## UNIT 26 PROBABILITY DISTRIBUTIONS-I

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### Structure

- 26.0 Objectives
- 26.1 Introduction
- 26.2 Elementary Concept of Random Variable
- 26.3 Probability Mass Function
- 26.4 Probability Density Function
- 26.5 Probability Distribution Functions
- 26.6 Moments and Moment Generating Functions
  - 26.6.1 Moments
  - 26.6.2 Moments Generating Functions
- 26.7 Let Us Sum Up
- 26.8 Key Words
- 26.9 Answers/Hints to Check Your Progress Exercises
- 26.10 Exercises

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### 26.0 OBJECTIVES

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After going through this unit, you should be able to

- discuss the concept of random variable;
- estimate the probability density function;
- find the probability mass function;
- distinguish between probability mass function and probability density function; and
- identify moments generating functions.

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### 26.1 INTRODUCTION

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The previous unit has discussed the theory of probability. This unit will introduce the concepts related to probability distribution. Random variables are always generated with a particular pattern of probability attached to them. Thus, based on the pattern of probability for the different values of random variable, we can distinguish them. Once we know these probability distributions and their properties, and if any random variable fits into a probability distribution, it will be possible to answer any question regarding the variable. In this unit, we have defined the random variable and made a broad distinction of the probability distributions based on whether the random variable is continuous or discrete. Then we have discussed how the moments of a probability distribution describe the distribution completely; how the technique of moment generating function could be used to obtain the moments of any probability distribution.

## 26.2 ELEMENTARY CONCEPT OF RANDOM VARIABLE

When a random experiment is performed, we are often not interested in the detailed results, but only in the value of some numerical quantity determined by the experiment. It is natural to refer to a quantity whose value is determined by the result of a random experiment as a random quantity.

A variable is called random when its occurrence depends on the chance factor. In other words, there is a probability that the variable will take a particular value. When we toss a coin, say thrice, the number of heads we obtain is a random variable. Suppose in this case,  $X$  denotes the number of heads. Clearly,  $X$  can take four values, 0, 1, 2, and 3. More importantly, there is a probability attached with every value of the variable  $X$ . Therefore,  $X$  is a random variable. We can cite several examples of random variable: the number of red cards drawn when we draw 10 cards from a pack of 52 cards, the number we obtain when a die is rolled, number of accidents in a city and the number of printing mistakes in a page of a newspaper.

Random variables could be either continuous or discrete. A discrete random variable takes only distinct values. In the above examples all are discrete variables. If  $Y$  is a variable that denotes the number when we roll a die, it will always take integral values. Practically, it will take any value among  $\{1, 2, 3, 4, 5, 6\}$  and cannot take the value 5.5 or 4.3. Therefore,  $Y$  is a discrete random variable.

Similarly, a random variable is continuous if it takes infinite number of values within a given interval or range. The life of an electrical gadget, duration of phone calls received by a telephone operator and the amount of annual rainfall in a particular district of India are examples of continuous random variables. A random variable always takes real values. Therefore, we can define a random variable as a real valued function defined over the points of a sample space (set of all possible outcomes of an experiment) with a probability measure.

## 26.3 PROBABILITY MASS FUNCTION

Probability distribution of a random variable is a statement specifying the set of its possible values together with the respective probabilities. When a random experiment is theoretically assumed to serve as a model, the probabilities could be represented as a function of the random variable. Let a discrete random variable  $X$  assume the values  $x_1, x_2, x_3 \dots x_n$  with probabilities  $p_1, p_2, p_3, \dots, p_n$ , satisfying the conditions:  $\sum_{i=1}^n p_i = 1, 0 \leq p \leq 1$ . The specification of the values of

$x_i$  together with the probabilities  $p_i$  defines the probability distribution of the random variable  $X$ . It is called the discrete probability distribution of  $X$ .

Often, the probability that the discrete random variable  $X$  assumes the value  $x$  is represented by  $f(x)$ , where,  $f(x) = P(X = x)$  = probability that  $X$  assumes a specific value  $x$ . The function  $f(x)$  is known as the probability mass function(p.m.f.).

The discrete random variable  $X$  can assume countably infinite number of values with the p.m.f.  $f(x)$ . The p.m.f.  $f(x)$  must satisfy the following two conditions:

- 1)  $f(x) \geq 0$
- 2)  $\sum_{i=1}^n f(x_i) = 1$

**Example:** An unbiased coin is tossed until the first head is obtained. If the random variable  $X$  denotes the number of tails preceding the first head, then what is the probability distribution of  $X$ ?

Here  $S = \{H, TH, TTH, TTTH, \dots\}$   $S = \{H, TH, TTH, TTTH, \dots\}$ .

|                                  |   |    |     |       |
|----------------------------------|---|----|-----|-------|
| Sample point                     | H | TH | TTH | ..... |
| X= No. of tails preceding a head | 0 | 1  | 2   |       |

The probability distribution of the random variable  $X$  is shown in the table. Clearly, the p.m.f. of  $X$  is  $F(X = x) = (1/2)^{n+1}$ . It satisfies the two conditions mentioned earlier, i.e.,  $f(x)$  is always greater than '0' and  $\sum_{i=1}^n f(x_i) = 1$ . Note that  $X$  is a countably infinite random variable.

**Table 26.1: Probability Distribution of X**

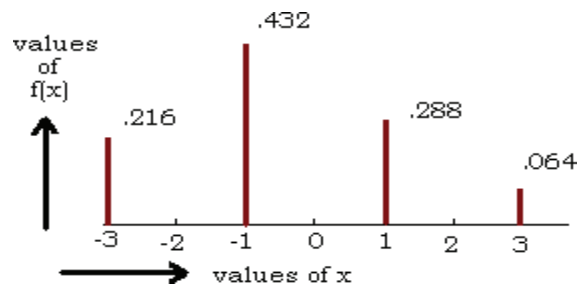
| Values of X | $f(X=x)$  |
|-------------|-----------|
| 0           | $(1/2)$   |
| 1           | $(1/2)^2$ |
| 2           | $(1/2)^3$ |
| 3           | $(1/2)^4$ |
| 4           | $(1/2)^5$ |
| -----       | ----      |
| -----       | ----      |
| N           | $(1/2)^n$ |

$$\begin{aligned}
 &= \sum_{i=1}^n f(x_i) = 1/2 + (1/2)^2 + (1/2)^3 + (1/2)^4 + \dots + (1/2)^n \\
 &= 1/2 \{ 1 + (1/2) + (1/2)^2 + (1/2)^3 + \dots + (1/2)^{n-1} \} \\
 &= 1/2 \cdot 1/(1-1/2) = \frac{1/2}{1/2} \\
 &= 1
 \end{aligned}$$

**Example:** Let  $X$  be a discrete random variable. Its probability distribution is given by Table 26.2. Its density could be represented by Fig. 26.1.

**Table 26.2: Probability distribution of  $X$**

| Values of $X$ | $f(X=x)$ |
|---------------|----------|
| -3            | 0.216    |
| -1            | 0.432    |
| 1             | 0.288    |
| 3             | 0.064    |



**Fig. 26.1: Probability density of  $X$**

## 26.4 PROBABILITY DENSITY FUNCTION

In the previous section, we discussed the probability distribution of a discrete random variable. But what if the random variable is continuous, i.e., it can take any value within a given range. Since the number of possible values, a continuous variable can take is uncountably infinite, we cannot assign a probability to each value taken by the variable, as we did in case of the discrete random variable. In case of continuous random variable, we assign probability to an interval, which is in the range of the relevant random variable. Therefore, a continuous random variable has its probability density function as  $f(x)$  if it is a continuous non-negative function and it gives the probability that the random variable will lie in a specified interval when integrated over the interval. Let  $x$  be a continuous random variable which can assume any value in the interval  $(a, b)$  i.e.,  $a \leq x \leq b$ .

Let  $f(x)$  be a non-negative function such that:

$$P(c \leq x \leq d) = \int_c^d f(x) dx$$

i.e., the probability of  $x$  lying in any given interval  $(c, d)$  can be obtained by integrating the function between the two limits of the interval.

The function  $f(x)$  is called the probability density function (p.d.f.) provided it satisfies the following two conditions

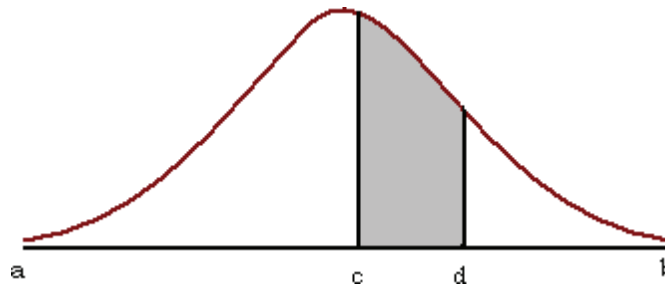
$$1) f(x) \geq 0$$

- 2) If the range of the continuous random variable is  $(a, b)$  such that  $\int_a^b f(x) dx = 1$ ;  
 $\int_a^b f(x) dx = 1$  where,  $a \leq x \leq b$  is the range of  $x$ .

$P(c \leq x \leq d) = \text{Area under the probability curve between the vertical line } c \text{ and } d \text{ (refer Fig. 26.2).}$

The shaded area in Fig. 26.2 represents the probability that the variable  $x$  will take values between the interval  $(c, d)$ , whereas its range is  $(a, b)$ . In the figure, we have taken the values of  $x$  in the horizontal axis and those of  $f(x)$  on the vertical axis. This is known as the probability curve. Since  $f(x)$  is a p.d.f., total area under the probability curve is 1 (i.e.,  $\int_a^b f(x) dx = 1$ ).

1) and the curve cannot lie below the horizontal axis as  $f(x)$  cannot take negative values. Any function that satisfies the above two conditions, can serve as a probability density function in the presence of real values only.



**Fig. 26.2: Probability curve**

**Example:** If  $x$  is a continuous random variable with the probability density function  $f(x) = k \cdot e^{-3x}$  for  $x > 0$  and 0 otherwise, then find  $k$  and  $P(0.5 \leq x \leq 1)$

If the function specified in the problem is to serve as a p.d.f., then the following two conditions must hold:

- 1)  $f(x) \geq 0$  for all values of  $x$
- 2) If the range of the continuous random variable is  $(a, b)$   $\int_a^b f(x) dx = 1$

Clearly, the range of the function given in the problem is  $-\infty$  to  $\infty$  and for every value of  $x$  within that interval,  $f(x)$  is positive, provided that 'k' is positive. To satisfy these condition.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\text{or, } \int_0^{+\infty} f(x) dx + \int_{-\infty}^0 f(x) dx = 1$$

$$\text{or, } \int_0^{+\infty} f(x) dx = 1 \int_0^{+\infty} f(x) dx = 1$$

$$\text{or } \int_0^{+\infty} k e^{-3x} dx = 1 \int_0^{+\infty} k e^{-3x} dx$$

$$\text{or, } k/3 = 1$$

$$\text{or, } k = 3$$

Thus, we get  $f(x) = 3e^{-3x}$  for  $x > 0$  and 0 otherwise.

$$P(0.5 \leq X \leq 1) = \int_{0.5}^1 3e^{-3x} dx = -e^{-3} + e^{-1.5} = 0.173$$

## 26.5 PROBABILITY DISTRIBUTION FUNCTION

The Cumulative distribution function (c.d.f) -  $F(x)$ , represents the probability that the random variable takes a value less than or equal to a specified value  $x$ . Thus  $F(x)$  shows the area under probability curve  $f(x)$  to the left of the ordinate at any specified value. Thus, for example:

$F(c) = P(x \leq c) = \int_{-\infty}^c f(x) dx$  i.e., Area under the probability curve to the left of the vertical line  $c$ .

$P(c \leq x \leq d) = P(c \leq x \leq d) = \text{Area between the ordinates } c \text{ and } d = (\text{Area to the left of the ordinate at } d \text{ minus } (\text{Area to the left of the ordinate at } c) = F(d) - F(c)$

Thus,  $P(c \leq x \leq c) = F(c) - F(c) = 0$  i.e.,  $P(x = c) = 0$ . Thus, for continuous probability distribution a random variable takes any specific value is zero.

The distribution function of a continuous random variable has the same nice properties as that of a discrete random variable viz.,

- i)  $F(-\infty) = 0, F(\infty) = 1$
- ii) If  $a < b$ , then  $F(a) \leq F(b)$  where  $a$  and  $b$  are any real number.
- iii) Furthermore, it follows directly from the definition that  $P(a \leq x \leq b) = F(b) - F(a)$ , where  $a$  and  $b$  are real constants with  $a \leq b$
- iv)  $F(x) = d/dx F(x)$  where the derivatives exist

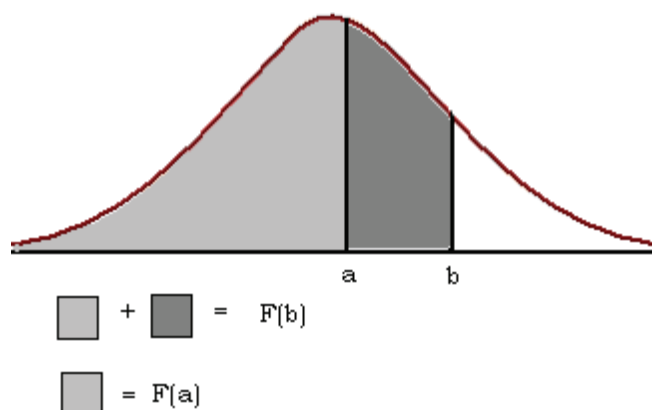


Fig. 26.3: Distribution of a Continuous Random Variable

**Example:** Find the distribution function of the random variable  $x$  of the previous example and use it to re-evaluate  $P(0.5 \leq x \leq 1)$

For all non-positive values of  $x$ ,  $f(x)$  takes the value 0. Therefore,

$$F(x) = 0 \text{ for } x \leq 0.$$

For  $x > 0$ ,

$$F(x) = \int_0^x 3e^{-3t} dt + \int_x^{\infty} f(t) dt$$

Thus,  $F(x) = 0$  for  $x \leq 0$ , and  $F(x) = 1 - e^{-3x}$  for  $x > 0$ .

To determine  $P(0.5 \leq x \leq 1)$ , we use the fourth property of the distribution function of a continuous random variable.

$$P(0.5 \leq x \leq 1) = F(1) - F(0.5) = (1 - e^{-3}) - (1 - e^{-3 \times 0.5}) = 0.173$$

## Check Your Progress 1

- 1) If  $X$  is a discrete random variable having the following p.m.f.,

|            |   |     |      |      |      |       |        |            |
|------------|---|-----|------|------|------|-------|--------|------------|
| $X$        | 0 | 1   | 2    | 3    | 4    | 5     | 6      | 7          |
| $P(X = x)$ | 0 | $k$ | $2k$ | $2k$ | $3k$ | $k^2$ | $2k^2$ | $7k^2 + k$ |

- determine the value of the constant  $k$
- find  $P(X < 5)$
- find  $P(X > 5)$

.....

.....

.....

.....

.....

- 2) For each of the following, determine whether the given function can serve as a p.m.f.

- $f(x) = (x - 2)/5$  for  $x = 1, 2, 3, 4, 5$
- $f(x) = x^2/30$  for  $x = 0, 1, 2, 3, 4$
- $f(x) = 1/5$  for  $x = 0, 1, 2, 3, 4, 5$

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## 26.6 MOMENTS AND MOMENTS GENERATING FUNCTIONS

### 26.6.1 Moments

Let  $X$  be a discrete random variable, which takes the values  $x_1, x_2, x_3, \dots, x_n$  and  $f(x)$  is the probability that  $X$  will take the value  $x$ . Then the expected value of  $X$  is given by

$$E(X) = \sum_{i=1}^n x_i f(x_i)$$

Correspondingly, if  $X$  is a continuous random variable and  $f(x)$  gives the probability density at  $x$ , the expected value of  $x$  is given by  $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

In the definition of continuous random variable, we assume that the integral exists, otherwise, the mathematical expectation is undefined. The expected value of a random variable gives its mean value if we think  $f(x)$  as the relative frequency of the random variable  $X$  when it takes the value  $x$ .

Again, if  $g(x)$  is a continuous function of the value of the random variable  $X$ , then the expected value of  $g(x)$  is given by  $E(g(x)) = \sum_{i=1}^n g(x_i) f(x_i)$ , when  $X$  is a discrete random variable. When it is a continuous random variable, the expected value of  $g(x)$  is given by  $E(g(x)) = \int_{-\infty}^{\infty} g(x) f(x) dx$ .

The determination of mathematical expectation can often be simplified by using the following theorems:

- 1)  $E(a + b.X) = a + b.E(X)$
- 2)  $E(a) = a$ , where  $a$  is a constant.
- 3) If  $c_1, c_2, c_3, \dots, c_n$  are constants,  $E\left[\sum_{i=1}^n c_i g(x_i)\right] = \sum_{i=1}^n c_i E[g(x_i)]$

In statistics as well as economics, the notion of mathematical expectation is very important. It is a special kind of moment. We now introduce the concept of moments and moment generating functions.

The  **$r^{\text{th}}$  order moment about the origin** of a random variable  $x$  is denoted by  $\mu'_r$  and is given by the expected value of  $x^r$ .

Symbolically, for  $x$  being a discrete random variable  $\mu'_r = E(x^r) = \sum_x x^r f(x)$ ,  $\mu'_r = E(x^r)$  for  $r = 1, 2, 3, \dots, n$

For  $x$  being a continuous random variable  $\mu'_r = E(x^r) = \int_{-\infty}^{\infty} x^r f(x) dx$

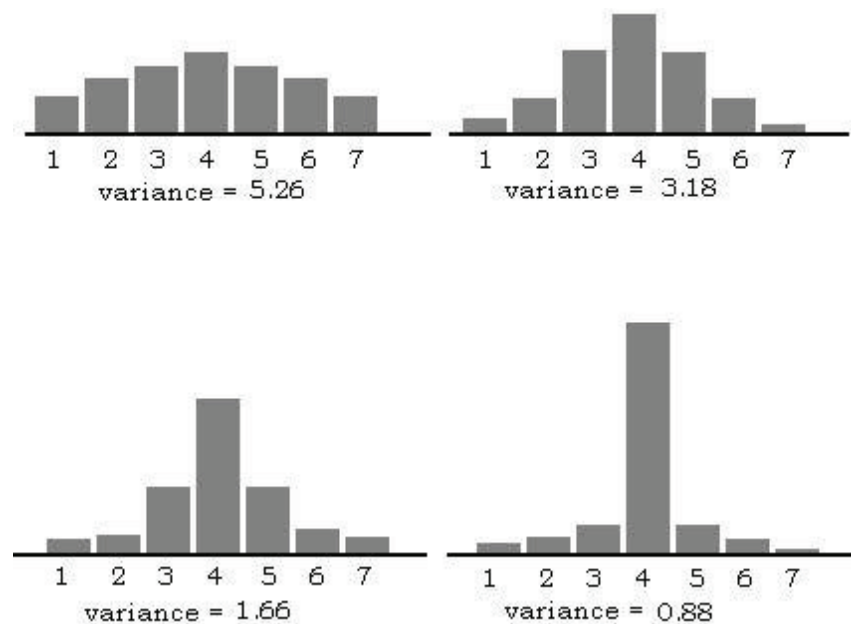
The term moment comes from physics. If  $f(x)$  symbolizes quantities of points of

masses, where  $x$  is discrete, acting perpendicularly on the  $x$  axis at distance  $x$  from the origin, then  $\mu'_1$  as defined earlier would give the center of gravity, which is the first moment about the origin. Similarly,  $\mu'_2$  gives the moments of inertia.

When  $r = 0$ , we have  $\mu'_0 = E(x^0) = E(1) = \int_{-\infty}^{+\infty} f(x) dx = 1$ . When  $r = 1$ , we have  $\mu'_1 = E(x^1) = E(x)$ . In statistics,  $\mu'_1$  gives the mean of a random variable and it is generally denoted by  $\mu$ .

The special moments we shall define are of importance in statistics because they are useful in defining the shape of the distribution of a random variable, viz., the shape of its probability distribution or its probability density.

The  $r^{\text{th}}$  **moment** of random variable about mean is given by  $\mu_r$ . It is the expected value of  $(X - \mu)^r$ , symbolically  $\mu^r = E((X - \mu)^r) = \sum_{i=1}^n (x_i - \mu)^r f(x_i)$ , for  $r=1,2,3,\dots,n$ , when  $x$  is discrete and continuous variable and when  $x$  is a continuous random variable,  $r^{\text{th}}$  moment is given by  $\mu_r = \int_{-\infty}^{\infty} (x - \mu)^r \cdot f(x) dx$ .



**Fig. 26.4: Distributions with same Means but different Variances**

Note that,  $\mu_0 = 1$  and  $\mu_1 = 0$  for any random variable for which  $\mu$  exists. The second moment about mean i.e.,  $\mu_2 = E(x - \mu)^2$  is called the variance and denoted by  $\sigma^2$  or  $\text{Var}(x)$ .

The above example shows how probability distributions vary with the variance of the random variable. A high value of  $\sigma^2$  means the spread of the distribution is thick at the tails and a low value of  $\sigma^2$  implies the spread of the distribution is tall at the mean of the distribution and the tails are flat. Similarly, third order